A Comparative Study of Deep Q-Networks and Traditional Algorithms for Multi-Agent Shortest-Path Search in Dynamic Graphs

Madhav Joshi

Marwadi University

Abstract

This study compares prioritized breadth-first search, Dijkstra and A* with solo-trained and shared-policy deep Q-networks for navigation on static and changing unit-cost grids. Ten policies from five training seeds were frozen after static 5×5 training and evaluated without learning on 120 new geometries spanning 5×5, 10×10 and 15×15 sizes, one/two/four agents and two wall probabilities. The final protocol produced 15,120 unique evaluation episodes; three fresh timing rollouts per episode were not treated as extra outcome samples. Across all conditions, replanning A* completed 642/720 episodes (89.17%), shared DQN 799/3600 (22.19%), and solo DQN 870/3600 (24.17%). All recorded first trajectories passed replay auditing, and actual collisions were zero under the common rejection rules. Success-conditioned paths, matched comparisons and descriptive variation across model seeds are reported alongside failures. The findings apply to fixed small-map-trained policies and current-map prioritized planners under this protocol; they do not establish a universal algorithm ranking or a general reinforcement-learning adaptation advantage.

Index terms: graph search; deep Q-network; multi-agent pathfinding; dynamic grid; policy transfer.

1. Introduction

Graph search supplies routes for software agents and mobile robots. Breadth-first search (BFS) finds minimum-edge paths in unweighted graphs [1], Dijkstra's algorithm addresses minimum-length graph paths [2], and A* directs search with a heuristic estimate [3]. These foundations solve an individual graph-search problem; using them for several simultaneous agents additionally requires temporal coordination. Multi-agent pathfinding (MAPF) has applications in automated warehouses and robotics, but collision and goal-occupancy assumptions differ between studies [6].

Reinforcement learning offers a different approach: a policy is learned before deployment and then reused to choose actions. DQN provides a neural approximation to action values [4]. A reusable policy might react to a changed neighborhood without running a fresh graph search, although that does not establish optimal paths, reliable generalization or a computational advantage over replanning. The present study tests these possibilities with a deliberately small method.

We compare fixed-priority space-time BFS, Dijkstra and A* with compact DQNs trained on static 5×5 maps. A solo-trained policy is deployed independently for each agent; a shared-policy model is trained with two simultaneous agents. Both are tested without further learning on fresh 5×5, 10×10 and 15×15 grids, one/two/four agents, and static or scheduled-changing obstacles. Frozen classical routes and untrained networks are controls. Our question is how these particular trained policies transfer, rather than whether reinforcement learning universally replaces classical search. The planners receive the full current graph and DQNs receive local features, so the comparison includes an information-access difference.

The study contributes (1) a tested environment with simultaneous proposals, absorbing goals and explicit dynamic snapshots; (2) a frozen, geometry-disjoint comparison retaining unsuccessful cases and five model seeds; and (3) an audited analysis of transfer, path quality, conflict attempts and runtime. The novelty is the controlled experiment and reproducible implementation, not a new MAPF or reinforcement-learning algorithm.

Before final collection we recorded four hypotheses. H1 predicts shorter static paths from A* than DQN. H2 predicts faster completion after obstacle changes from DQN than fixed precomputed routes, with replanning also evaluated. H3 predicts lower success and more conflict attempts with more agents. H4 predicts stronger success from shared training than independent use of a solo-trained policy. Each is assessed descriptively on matched cases; contrary evidence is retained.

2. Related Work

2.1 Classical shortest-path algorithms

BFS uses graph-distance levels [1]. Dijkstra searches minimum-length paths [2], while A* adds heuristic guidance [3]. In our unit-cost grid, Manhattan distance is the heuristic. These single-agent foundations do not make the entire fixed-priority team planner optimal. Our team searches add a time coordinate, wait actions and reservations, following the cooperative-search principle [5].

2.2 Deep reinforcement learning and DQN

Mnih et al. demonstrated learned control through neural action values [4]. Our implementation uses the DQN ideas of experience replay and a target network, with a small multilayer perceptron receiving a vector of navigation features. It differs from pixel-based Atari control in architecture, rewards and simulator assumptions. We examine trained policies' failures as well as successes.

2.3 Reinforcement learning on graphs

Dai and Khalil et al. combine reinforcement learning with graph embeddings for combinatorial optimization [9]. That work concerns objectives such as vertex cover, graph cuts and traveling-salesman solutions. Our DQN uses no graph embedding and chooses local navigation actions. Consequently, graph-optimization results cannot be imported as evidence for collision-safe pathfinding here.

2.4 Multi-agent pathfinding

Silver's cooperative search uses space-time routes and reservations [5]. Stern et al. clarify MAPF variants and evaluation conventions [6]. Learned MAPF methods can be substantially richer than our baseline: PRIMAL combines reinforcement and expert imitation learning [7], and Li et al. communicate local features through graph neural networks trained by expert imitation [8]. We use neither expert demonstrations nor communication. Their results therefore motivate future work rather than provide performance numbers comparable to our maps.

Source	Problem/method	Setting	Difference from this experiment
Dijkstra [2], Hart et al. [3]	Minimum-cost graph search	Known graphs	No learned simultaneous-agent policy
Mnih et al. [4]	DQN control	Atari	Different observation/action domain
Silver [5]	Cooperative space-time search	Game navigation	Includes richer cooperative search variants
Stern et al. [6]	MAPF terminology/benchmarks	Grid MAPF	Our custom maps are not their benchmark
Sartoretti et al. [7]	RL with imitation	Decentralized MAPF	Our small DQN lacks imitation learning
Li et al. [8]	GNN feature communication	Multi-robot planning	Our agents do not communicate
Dai/Khalil et al. [9]	RL with graph embeddings	Combinatorial graph tasks	Different objective and representation

The selected experimental gap is a transparent comparison of small static-trained local DQNs and simple classical replanning under matched graph changes. This is not a claim that such comparisons have never been studied.

3. Problem Formulation

At action tick t the environment has a unit-cost grid graph G_t = (V_t, E_t). Free cells are vertices and orthogonal free-cell adjacency defines movement edges. There are n agents, with positions p_t^i, distinct starts s^i and distinct goals g^i. Each may move up, down, left or right, or wait. Upon first arrival an agent remains at its goal, continuing to occupy that vertex. The global state includes positions, goals, the current graph and elapsed time; a policy receives only its feature observation. The local observation is generally partially observable, so we do not assert that its vector is a Markov state.

All actions are proposed simultaneously. A moving proposal into an obstacle or outside the grid fails. Moving proposals sharing a destination and opposite edge swaps are rejected. Rejection then propagates until a move cannot enter a cell whose occupant now stays. Following a cell vacated in the same tick and cycles longer than two are permitted. This centralized movement referee enforces collision safety for every method, without supplying extra information to the

learned policies. Rejected-agent attempts and actual collisions are distinct.

Wall changes prepare before the corresponding action observation. At tick 2 a safe free cell is added to the obstacle set, then removed at tick 8. All eight candidate team endpoints are protected, and the added cell is initially at Manhattan distance at least 2 from all four starts. No agent can occupy it after only one move. Events are deterministic and shared across methods and nested team sizes, but completion stops subsequent events. Empty schedules remain in the denominator if no eligible cell exists. The route-dependent perturbation need not obstruct every selected method's path.

Our primary objective is feasible team completion within horizon H. We then report makespan T = max_i T_i, sum of first-arrival times sum_i T_i, executed movement count L, conflict attempts and computational time. A single weighted objective is not minimized by all implementations: prioritized planners minimize each agent's arrival turns in fixed order, and DQN optimizes a shaped return. This distinction prevents a misleading claim of a shared global optimality target.

4. Methodology

4.1 Environment and observations

The v2 observation has 17 normalized features: current row/column (2), signed goal displacement (2), neighboring wall flags (4), neighboring boundary flags (4), remaining time (1), and adjacent other-agent occupancy flags (4). Coordinates and goal displacements are normalized by each grid's row/column span. Policy inputs contain these features only, with no future schedule, full graph or diagnostic counters. All final one-agent cases also use this v2 team interface; earlier 13-feature development results are not pooled into final comparisons.

The observation/action/reward loop is illustrated in Figure 1. After joint movement, a fresh next observation includes any event prepared for the next action tick. Reset restores the original graph, positions and event history.

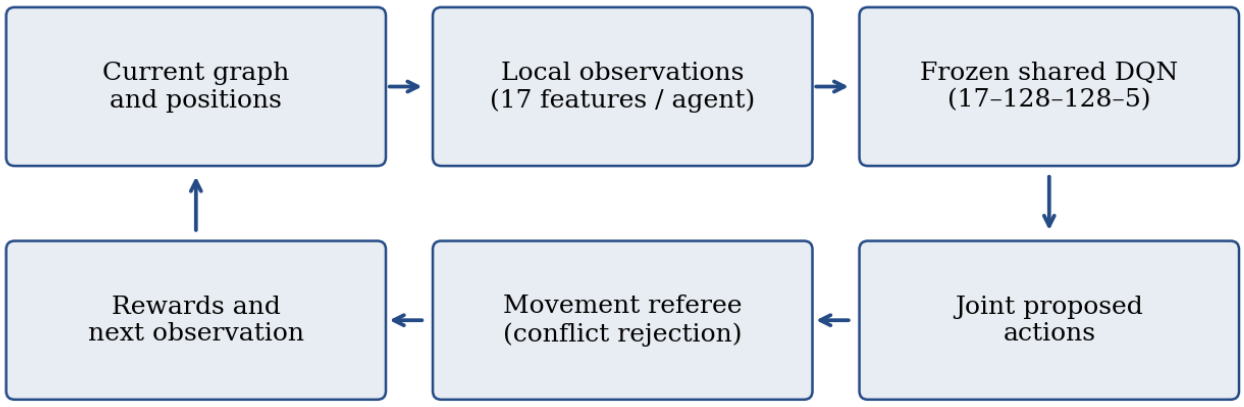


Figure 1. Shared-policy evaluation loop: current graph and positions produce local observations; frozen DQN weights propose actions; the movement referee resolves conflicts, then returns named rewards and the next observation.

4.2 Rewards and finite-horizon learning

An active agent receives a -1 turn cost; additional penalties are -10 for an obstacle attempt, -5 for a boundary attempt, and -25 for a rejected conflict attempt. First goal arrival awards +100 once. Team completion awards +100 once to every agent, including agents that arrived earlier. Finished agents incur no turn cost. Reset-at-goal gives no free completion bonus. Named components are logged separately. The reward is therefore the sum of applicable components, not mutually exclusive event labels in a table.

The online network estimates Q(o,a;theta). For replay samples the target is

$$y = r + \gamma(1 - d)\max_{a'} Q(o', a'; \theta^{-}),$$

where d is team termination or finite-horizon truncation. The loss is

$$\mathcal{L}(\theta) = \frac{1}{B}\sum_{j=1}^{B}[y_j - Q(o_j, a_j; \theta)]^2.$$

Both endings mask the bootstrap because there is no continuation beyond this task's allowed horizon. Shared replay retains arrived-agent forced-WAIT transitions until the team ends, preserving their delayed team bonuses.

4.3 Models and training

The network is 17–128–128–5, with ReLU hidden activations. Adam uses learning rate .001, discount .99, batch 64, replay capacity 50,000, warm-up 256 transitions, an update every eight transitions, target copying every 500 optimizer updates, and gradient clipping at norm 10. Epsilon falls linearly from 1 to .05 over 600 cycles. Five training seeds (0–4) are retained.

Models train on 50 static 5×5 geometries at .10 wall probability; ten validation geometries are observed without best-checkpoint selection. Each seed/mode uses 1,000 cycles. A solo cycle consists of two one-agent tasks; a shared cycle consists of one two-agent task. Both assign 2,000 agent tasks per seed. They do not match realized transitions, updates or reward timing: solo team bonuses arrive with individual completion, while shared bonuses wait for the last agent. These differences are reported as limitations of H4's causal interpretation. Training geometry generation also kept the two opposite corner cells free, before sampling team endpoints from all free cells; final wall sampling covers every cell. Solo training has zero other-agent occupancy features, whereas shared training encounters nearby teammates. Matching vector formats therefore does not match the full observation distribution at deployment.

Final weights are retained for every seed. During evaluation epsilon is zero, gradients and learning are disabled, and weights/update counters are checked for preservation. There is no online obstacle-change adaptation through training. An untrained v2 network for each seed is a diagnostic control. Parameter sharing does not give agents explicit communication or a centralized critic.

4.4 Classical methods and controls

BFS uses a FIFO queue; Dijkstra and A* use stable heaps. Space-time states are (cell, tick), with unit turn costs including waits. Reservations forbid occupied vertices, opposite edges and permanent earlier-agent goals. Priority is agent index. Plans use the current graph and remaining horizon, and are recomputed after graph changes or execution deviations. A failed complete plan leads to joint waiting until a relevant change or timeout, rather than immediate episode removal. This decoupled planner is not complete or globally team-optimal. The same space-time planner is used for one-agent final cases, including wait states. Its runtime is therefore not the runtime of an optimized spatial-only static shortest-path implementation, even though successful static one-agent paths can be checked against the exact spatial BFS distance.

Frozen variants compute their routes only on the initial graph. Each agent advances its route cursor only when the next route cell is executed; blocked movements are retried. Resulting delays may invalidate original temporal reservations, although the common referee still rejects conflicts. Frozen routes are controls, not the main comparison for a changing graph.

5. Experimental Setup

5.1 Frozen evaluation design

The protocol was fixed before generating final cases. There are three grid sizes (5, 10, 15), two wall probabilities (.10, .20), two endpoint arrangements (uniform, crossing), three nested agent counts (1, 2, 4) and two conditions (static, dynamic). Horizons are 30, 60 and 90 turns by size. Ten final geometries per size/density/arrangement yield 120 geometries. Six classical variants and fifteen policy/seed pairs yield 21 runs per agent/condition case: 15,120 unique episodes. Each final episode has three fresh deterministic timing rollouts, 45,360 rollouts total. Only repeat zero contributes an outcome/trajectory.

Wall sampling includes every cell. Endpoints are sampled from free cells with RNG seed map_seed + 70000. Uniform sampling draws eight distinct cells, four starts followed by four goals. Crossing sampling draws four free left-edge starts

ordered top-to-bottom and four free right-edge goals ordered bottom-to-top. The first one/two pairs form the smaller-team tasks. Crossing is a traffic proxy and also changes route length; it does not guarantee high congestion. Initial deterministic BFS-route overlap pairs are recorded descriptively.

Final seed intervals start at 200000 and advance by 10000 per stratum in size/density/arrangement order. Duplicate geometries, too few free cells or too few free boundary endpoints for crossing are rejected and logged. Unreachable tasks are retained. Final geometry excludes all recorded development and pilot geometries, with 280 exclusions fixed in the seal. Requested probability and realized density are distinct. Pilot runs validate software and are excluded from all final research tables.

5.2 Metrics and analysis

Success requires all agents to arrive within the horizon. Failures remain in success denominators; completed makespan and arrival sum are missing on failures. Path and completion means use successful episodes and report sample counts. The exact relative movement gap $(L-L^*)/L^*$ is computed only for successful static one-agent episodes with positive reference distance. Independent shortest paths give a team lower bound, not a global feasible optimum. Dynamic/team gaps are left missing with reasons rather than fabricated.

Learned results report seed rates and their sample standard deviation on shared maps. These are descriptive model variability, not confidence intervals or independent map replications. Matched shared-versus-solo comparisons preserve map, agent count, condition and seed. Classical paired comparisons have no training seed. Learned-to-classical pairs reuse a classical reference for each model seed, without pretending the reference has independently trained replicas. Completion comparisons use jointly successful cases and state their denominator.

For H2 we separately record whether an addition was observed. Successful exposed completion steps are $T - 2 + 1$; a failure does not acquire a made-up completion time. Both-exposed paired comparisons also report success differences. Conditioning on exposure or joint success can change the sampled difficulty, so such comparisons alone do not establish an adaptation advantage.

5.3 Hardware, timing and validation

Execution uses a local Windows 11 CPU system, Python 3.12.6, PyTorch 2.11.0, NumPy 1.26.4 and Matplotlib 3.10.0, with eight logical CPUs reported by the environment and one PyTorch thread. Exact processor/platform strings are archived in run metadata; no unsupported hardware model or GPU is assumed. Repeated planning, action-selection and complete-episode timers are stored separately. Training timers come from the saved training logs. Method order is fixed and there is no timing warm-up; machine load and logging overhead limit speed claims. Recorded training seconds sum timed episode loops, including optimizer work; they exclude validation, setup and checkpoint saving and are not an end-to-end training-job wall time. Expanded classical states count processed space-time nodes, including popped goals and excluding stale heap entries, summed over initial planning and replans. The corresponding metric is not defined for neural action selection.

Correctness tests cover known/weighted shortest paths, simultaneous collision resolution, goal absorption, dynamic visibility, failed plans, model versions, replay targets and metric denominators. The saved final audit regenerates scenarios and re-executes every first-trajectory policy/planner decision, checking observations, rewards, plans, metrics, hashes and common active-event prefixes. Timing repetitions are checked for identical non-time outcomes during collection; the audit validates repetition counts and finite nonnegative timers, rather than claiming to reproduce machine elapsed times exactly. A recorded success is never substituted for this evidence.

6. Results

6.1 Collection, validation and sample denominators

The sealed experiment completed 15,120 unique episodes and 45,360 fresh timed rollouts on 120 geometries. The saved audit replayed 652,244 first-trajectory steps with zero actual collisions. Empty dynamic schedules numbered 0; any such cases remained in the denominator. Outcomes count only the first rollout per case; policy seeds reuse the same scenario sets. Each classical variant contributes 720 episodes, while each policy type contributes 3,600 across five models. These unequal totals represent equivalent matched scenarios, not extra distinct geometries for DQN.

Initially unreachable individual tasks occurred on 7, 9, and 16 of the 120 geometries for the nested one-, two-, and four-agent sets respectively. Such cases remain failures when they cannot be completed; reachability alone does not establish full team feasibility. Generation rejected 18 candidates according to the predeclared geometry/free-endpoint rules, with reasons recorded in the bank.

6.2 Success, agent scaling and map-size transfer

Table 1 summarizes all evaluated conditions, including frozen and untrained controls. Classical rows have no training-seed standard deviation. Learned seed standard deviations describe five frozen models on the same maps; they are not confidence intervals. Successful moves/makespan can favor a method that finishes only easy cases, so Table 1 must be read with its success denominators.

Table 1. Overall final outcomes; path/time means use successful episodes.

Method	Successes / episodes	Success (%)	Seed SD (pp)	Successful moves	Successful makespan	Mean attempts
astar	642/720	89.17	n/a	24.433	12.902	0.000
astar_frozen	612/720	85.00	n/a	22.882	14.067	5.624
bfs	644/720	89.44	n/a	24.646	12.919	0.000
bfs_frozen	613/720	85.14	n/a	23.021	14.119	5.957
dijkstra	644/720	89.44	n/a	24.646	12.919	0.000
dijkstra_frozen	613/720	85.14	n/a	23.021	14.119	5.957
shared	799/3600	22.19	5.173	11.436	9.947	4.522
solo	870/3600	24.17	3.082	12.875	10.775	11.723
untrained	15/3600	0.42	0.481	3.200	12.667	7.612

BFS completed 644/720 episodes and Dijkstra completed 644/720; A* completed 642/720. Equal-cost BFS and Dijkstra use equivalent arrival-time objectives here. Any difference from A* team outcomes does not contradict admissible single-agent A*: fixed priorities and route choices can change later-agent feasibility. This experiment does not isolate tie-breaking as a causal factor.

Table 2 separates static and dynamic agent-count conditions. The full 648-stratum supplement preserves density, grid size, endpoint arrangement and agent count together. Figure 2 displays success and Figure 3 completed makespan; methods with no successes have missing completion means.

Table 2. Final outcomes by environment condition and agent count.

Method	Condition	Agents	Successes / episodes	Success (%)	Successful makespan	Mean attempts
astar	dynamic	1	113/120	94.17	12.283	0.000
astar	dynamic	2	111/120	92.50	13.144	0.000
astar	dynamic	4	97/120	80.83	15.134	0.000
astar	static	1	113/120	94.17	10.991	0.000
astar	static	2	111/120	92.50	12.072	0.000
astar	static	4	97/120	80.83	14.289	0.000
astar_frozen	dynamic	1	113/120	94.17	15.363	0.000
astar_frozen	dynamic	2	102/120	85.00	16.000	7.100
astar_frozen	dynamic	4	76/120	63.33	16.750	26.642
astar_frozen	static	1	113/120	94.17	10.991	0.000
astar_frozen	static	2	111/120	92.50	12.072	0.000
astar_frozen	static	4	97/120	80.83	14.289	0.000
bfs	dynamic	1	113/120	94.17	12.336	0.000
bfs	dynamic	2	111/120	92.50	13.180	0.000
bfs	dynamic	4	98/120	81.67	15.133	0.000
bfs	static	1	113/120	94.17	10.991	0.000
bfs	static	2	111/120	92.50	12.072	0.000
bfs	static	4	98/120	81.67	14.265	0.000
bfs_frozen	dynamic	1	113/120	94.17	15.496	0.000
bfs_frozen	dynamic	2	103/120	85.83	16.097	6.833
bfs_frozen	dynamic	4	75/120	62.50	16.880	28.908
bfs_frozen	static	1	113/120	94.17	10.991	0.000
bfs_frozen	static	2	111/120	92.50	12.072	0.000
bfs_frozen	static	4	98/120	81.67	14.265	0.000
dijkstra	dynamic	1	113/120	94.17	12.336	0.000
dijkstra	dynamic	2	111/120	92.50	13.180	0.000
dijkstra	dynamic	4	98/120	81.67	15.133	0.000
dijkstra	static	1	113/120	94.17	10.991	0.000
dijkstra	static	2	111/120	92.50	12.072	0.000
dijkstra	static	4	98/120	81.67	14.265	0.000
dijkstra_frozen	dynamic	1	113/120	94.17	15.496	0.000
dijkstra_frozen	dynamic	2	103/120	85.83	16.097	6.833
dijkstra_frozen	dynamic	4	75/120	62.50	16.880	28.908
dijkstra_frozen	static	1	113/120	94.17	10.991	0.000
dijkstra_frozen	static	2	111/120	92.50	12.072	0.000
dijkstra_frozen	static	4	98/120	81.67	14.265	0.000
shared	dynamic	1	242/600	40.33	10.831	0.000
shared	dynamic	2	126/600	21.00	9.849	1.482
shared	dynamic	4	37/600	6.17	11.027	12.287
shared	static	1	238/600	39.67	9.647	0.000

Method	Condition	Agents	Successes / episodes	Success (%)	Successful makespan	Mean attempts
shared	static	2	120/600	20.00	8.625	1.503
shared	static	4	36/600	6.00	9.639	11.860
solo	dynamic	1	254/600	42.33	10.106	0.000
solo	dynamic	2	155/600	25.83	12.316	5.243
solo	dynamic	4	31/600	5.17	17.774	29.948
solo	static	1	249/600	41.50	8.839	0.000
solo	static	2	149/600	24.83	10.832	5.343
solo	static	4	32/600	5.33	16.625	29.802
untrained	dynamic	1	6/600	1.00	14.667	0.000
untrained	dynamic	2	1/600	0.17	12.000	4.043
untrained	dynamic	4	0/600	0.00	n/a	18.910
untrained	static	1	7/600	1.17	11.143	0.000
untrained	static	2	1/600	0.17	12.000	4.142
untrained	static	4	0/600	0.00	n/a	18.575

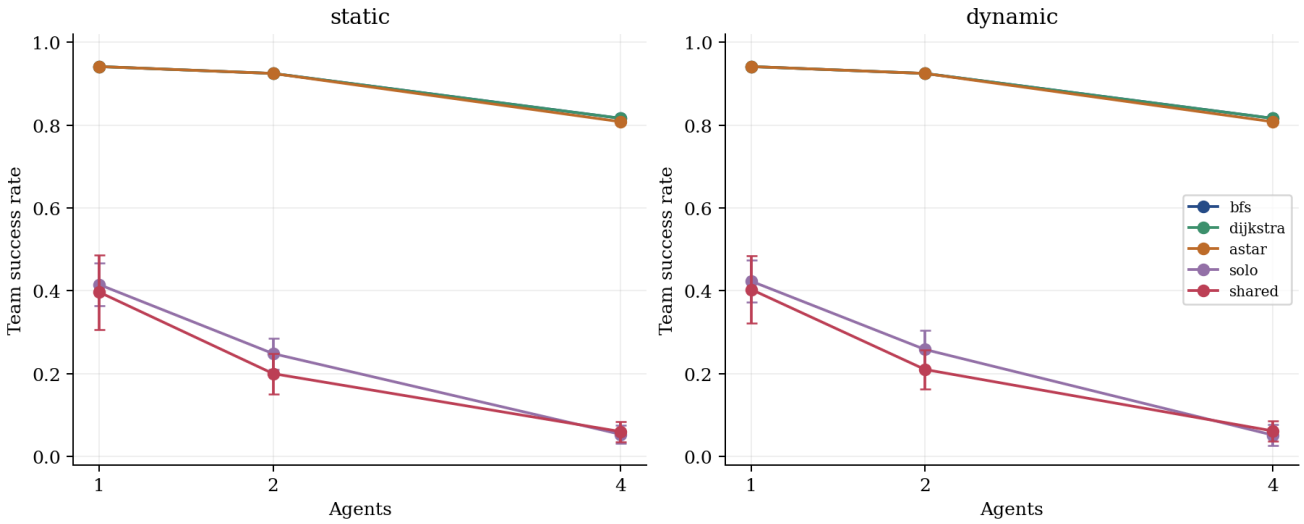


Figure 2. Success by agent count. Error bars for solo/shared show sample standard deviation across five model seeds; all failed episodes remain in denominators.

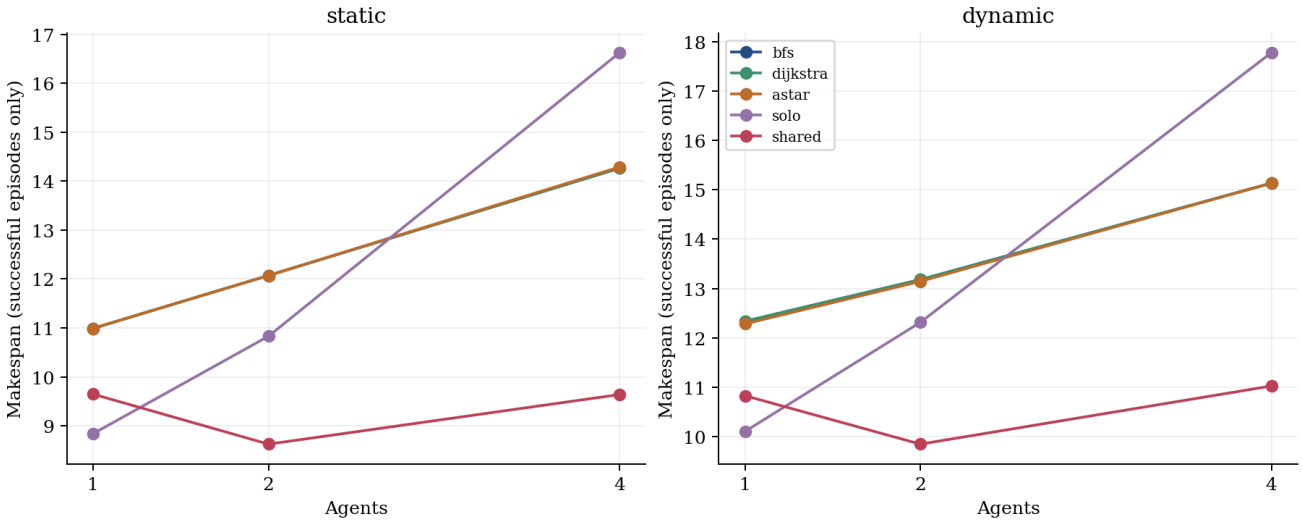


Figure 3. Successful-episode makespan by agent count. These curves condition on different successful subsets and do not alone show paired superiority.

Table 3. Map-size transfer, pooled across agent counts and endpoint arrangements.

Method	Condition	Grid side	Successes / episodes	Success (%)	Seed SD (pp)
astar	dynamic	10	110/120	91.67	n/a
astar	dynamic	15	114/120	95.00	n/a
astar	dynamic	5	97/120	80.83	n/a
astar	static	10	110/120	91.67	n/a
astar	static	15	114/120	95.00	n/a
astar	static	5	97/120	80.83	n/a
bfs	dynamic	10	110/120	91.67	n/a
bfs	dynamic	15	114/120	95.00	n/a
bfs	dynamic	5	98/120	81.67	n/a
bfs	static	10	110/120	91.67	n/a
bfs	static	15	114/120	95.00	n/a
bfs	static	5	98/120	81.67	n/a
dijkstra	dynamic	10	110/120	91.67	n/a
dijkstra	dynamic	15	114/120	95.00	n/a
dijkstra	dynamic	5	98/120	81.67	n/a
dijkstra	static	10	110/120	91.67	n/a
dijkstra	static	15	114/120	95.00	n/a
dijkstra	static	5	98/120	81.67	n/a
shared	dynamic	10	125/600	20.83	5.621
shared	dynamic	15	94/600	15.67	7.250
shared	dynamic	5	186/600	31.00	6.163
shared	static	10	122/600	20.33	5.224
shared	static	15	91/600	15.17	8.258
shared	static	5	181/600	30.17	5.786
solo	dynamic	10	134/600	22.33	3.970
solo	dynamic	15	120/600	20.00	7.862
solo	dynamic	5	186/600	31.00	2.236
solo	static	10	134/600	22.33	3.302
solo	static	15	113/600	18.83	7.919
solo	static	5	183/600	30.50	1.918

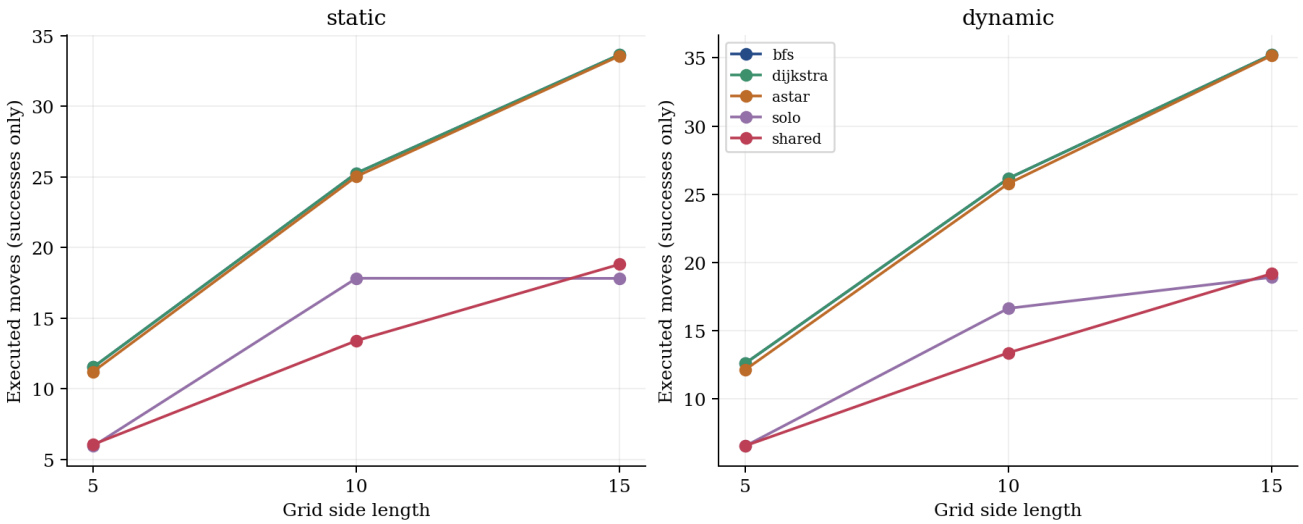


Figure 4. Executed movement count by grid size among successful episodes. Success rates and matched comparisons are required to interpret these conditional averages.

6.3 Matched static paths and dynamic completion

For H1, Table 4 compares each learned run with A* on exactly the same static task. Only jointly successful pairs contribute path differences; the same deterministic A* reference is reused for each learned seed. A positive movement difference means the learned policy used more moves. Team A* is prioritized arrival-time search, not a globally optimal movement reference.

Table 4. Learned-policy minus A* static movements on jointly successful pairs.

Policy minus A*	Agents	Pairs	Joint success	Move difference	Policy shorter / ties / A* shorter
solo	1	600	249	1.462	0 / 233 / 16
solo	2	600	149	2.940	0 / 132 / 17
solo	4	600	32	5.188	0 / 22 / 10
shared	1	600	238	2.185	0 / 218 / 20
shared	2	600	120	1.283	1 / 107 / 12
shared	4	600	36	0.889	0 / 26 / 10

The exact single-agent relative movement gap is additionally reported in Table 5. Its denominator is the exact spatial BFS distance, and its mean includes only successful static single-agent episodes. An absent gap for a failed episode is not treated as zero. Multi-agent and dynamic gaps remain undefined rather than using a misleading global optimum.

Table 5. Static single-agent exact relative movement gaps, (L - L*) / L*.

Method	Successes / episodes	Mean exact movement gap
astar	113/120	0.000
astar_frozen	113/120	0.000
bfs	113/120	0.000
bfs_frozen	113/120	0.000
dijkstra	113/120	0.000
dijkstra_frozen	113/120	0.000
shared	238/600	0.306
solo	249/600	0.307
untrained	7/600	0.000

For H2, Table 6 restricts pairs to episodes in which both methods observed the scheduled addition. A positive completion-step difference means slower learned completion, and a negative success difference means lower learned success in that exposed subset. Both-exposed cases and jointly successful cases have different denominators; neither excludes failed episodes from the reported paired success difference. Replanning controls remain the main fair classical methods.

Table 6. Learned-policy minus frozen A* under dynamic addition exposure.

Policy minus frozen A*	Agents	Both exposed pairs	Joint success	Success difference (pp)	Post-addition step difference
solo	1	600	254	-51.833	-1.374
solo	2	600	155	-59.167	0.723
solo	4	600	30	-58.167	4.333
shared	1	600	242	-53.833	-0.636
shared	2	600	121	-64.000	-0.669
shared	4	600	33	-57.167	0.212

6.4 Attempts, shared training and computational measurements

Figure 5 reports rejected-agent attempts per episode, including failures. These attempts indicate conflicting proposals; they are not realized physical collisions. Actual collisions were zero for every method because the common environment rejected unsafe proposals.

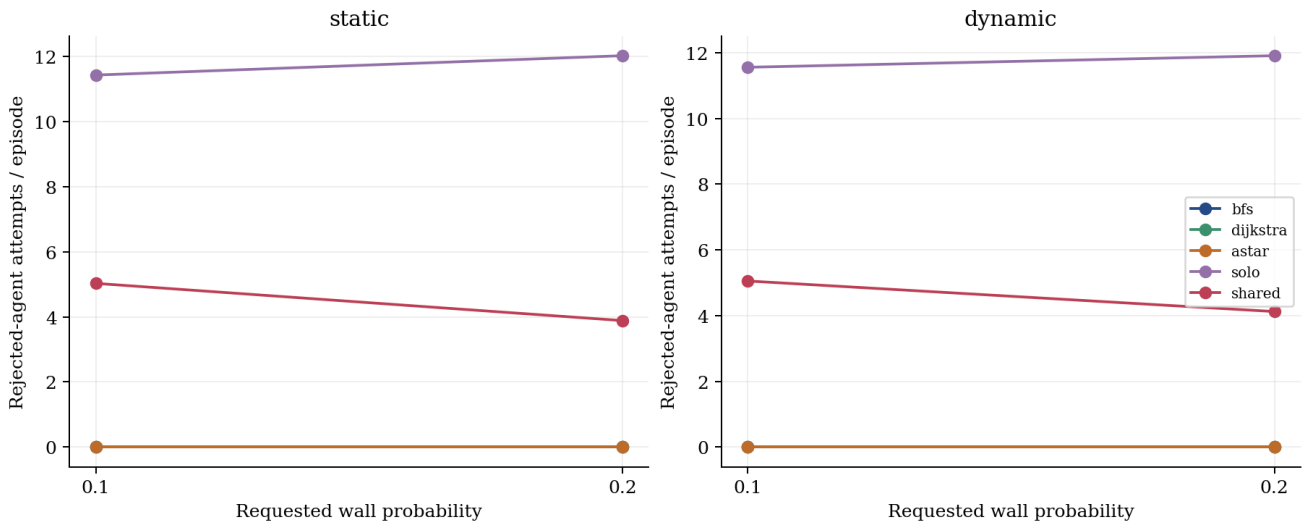


Figure 5. Conflict attempts by requested wall probability, averaged over all episodes. These are rejected-agent counts per turn summed over a trip; actual collision count is zero.

Table 7 pairs shared and solo policies using identical scenario and seed keys. Positive success differences favor shared training; path differences include only jointly successful pairs. Additional per-stratum paired differences are included in the machine-readable analysis.

Table 7. Shared-policy minus solo-policy matched effects.

Condition	Agents	Pairs	Joint success	Shared minus solo success (pp)	Paired seed SD (pp)	Move difference
static	1	600	184	-1.833	11.356	-0.370
static	2	600	89	-4.833	6.882	-1.258
static	4	600	14	0.667	2.528	-3.714
dynamic	1	600	190	-2.000	11.205	-0.274
dynamic	2	600	95	-4.833	7.715	-0.147
dynamic	4	600	13	1.000	3.084	-2.769

Table 8 separates recorded training budgets from inference. Task assignments were matched, while actual optimizer updates, transitions and reward timing were not. Figure 6 shows exploratory training returns from the original frozen-model training run; the curves are practice data, not final test results.

Table 8. Actual CPU training totals across five seeds per mode.

Mode	Tasks	Episodes	Transitions	Updates	Recorded seconds
solo	10000	10000	81770	10064	52.112
shared	10000	5000	121408	15020	80.586

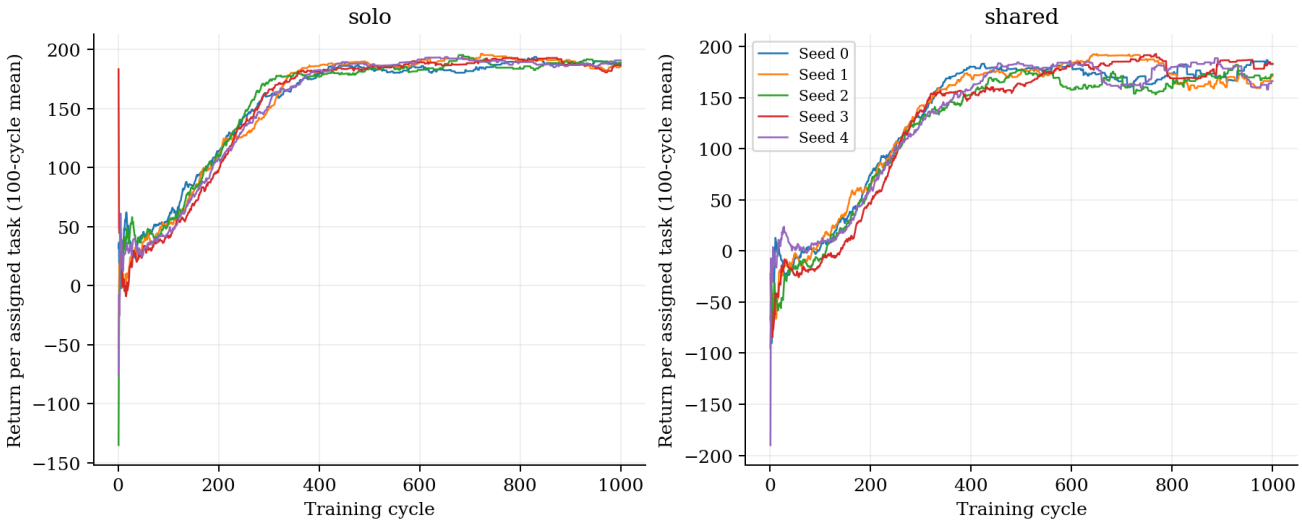


Figure 6. Training return per assigned task, trailing 100-cycle averages, for the frozen final solo/shared models. Curves are historical training data and do not select test checkpoints.

Table 9 uses per-case means from three fresh timing rollouts, averaged across static and dynamic cases. Planning, action-selection and complete-episode timers are separate; expanded space-time states sum over planning calls per episode and are missing for neural policies. Different success/horizon behavior changes how many decisions occur, so per-episode totals cannot establish a universal computational ranking. Figure 7 combines planning and selection only and retains those limitations.

Table 9. Repeated timing means in milliseconds and mean expanded states per episode.

Method	Agents	Cases	Planning ms	Selection ms	Total ms	Expanded states
astar	1	240	3.826	0.224	6.965	166.050
astar	2	240	9.775	0.362	14.861	678.654
astar	4	240	19.145	0.488	28.457	1362.312
astar_frozen	1	240	1.868	0.128	5.271	87.775
astar_frozen	2	240	4.711	0.187	10.423	357.075
astar_frozen	4	240	10.213	0.300	21.854	728.633
bfs	1	240	16.854	0.264	20.471	1066.158
bfs	2	240	31.516	0.383	36.641	2242.467
bfs	4	240	64.762	0.528	74.107	4518.108
bfs_frozen	1	240	8.832	0.129	12.409	600.392
bfs_frozen	2	240	17.938	0.190	23.633	1261.025
bfs_frozen	4	240	37.946	0.339	49.870	2554.758
dijkstra	1	240	17.851	0.243	21.212	1066.158
dijkstra	2	240	33.985	0.373	39.118	2242.467
dijkstra	4	240	70.675	0.517	79.841	4518.108
dijkstra_frozen	1	240	10.208	0.138	13.733	600.392
dijkstra_frozen	2	240	19.996	0.201	25.809	1261.025
dijkstra_frozen	4	240	41.336	0.311	53.417	2554.758
shared	1	1200	0.000	23.195	33.474	n/a
shared	2	1200	0.000	46.158	63.238	n/a
shared	4	1200	0.000	96.614	123.251	n/a
solo	1	1200	0.000	22.335	32.220	n/a
solo	2	1200	0.000	43.423	59.408	n/a
solo	4	1200	0.000	94.625	121.429	n/a
untrained	1	1200	0.000	35.191	50.452	n/a
untrained	2	1200	0.000	69.653	90.214	n/a
untrained	4	1200	0.000	131.137	158.751	n/a

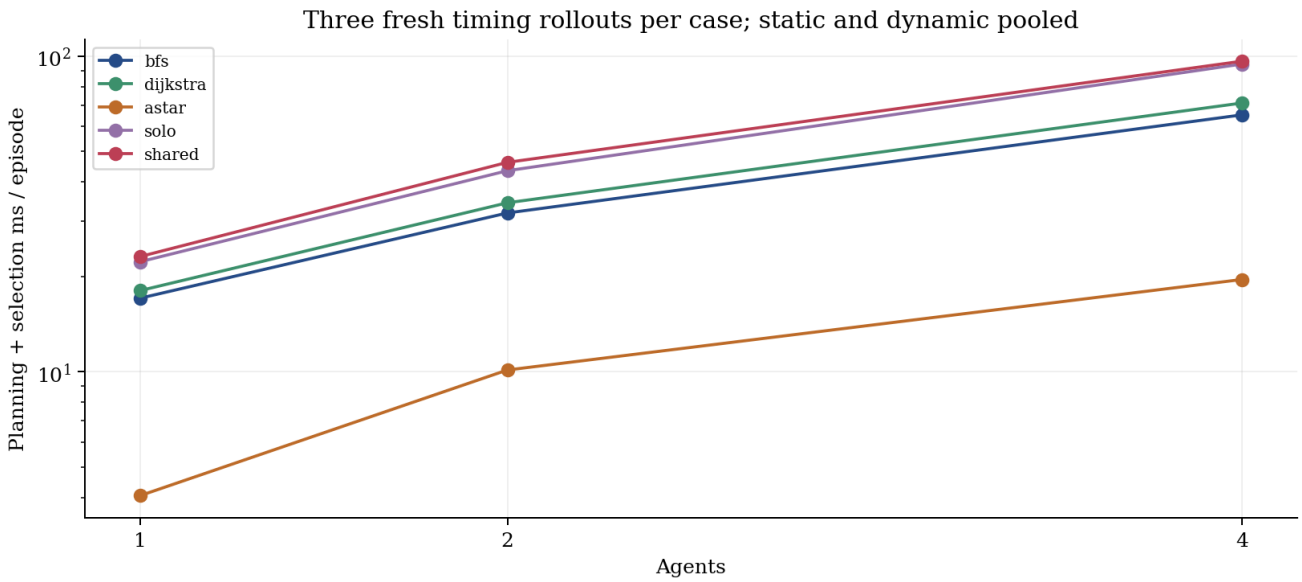


Figure 7. Planning plus action-selection time per episode, with three fresh rollouts per unique case. Fixed method order, different completion lengths and logging overhead limit runtime claims.

7. Discussion

Replanning A* succeeded on 89.17% of its evaluated cases. Shared and solo DQN reached 22.19% and 24.17% respectively. These averages combine conditions and should not replace the stratified tables. The main research question is answered for these implementations: the results quantify the reliability and path/timing tradeoffs of static-small-map-trained policies against current-map prioritized search. They are not a statement about the best attainable DQN, all reinforcement learning, or all MAPF planners.

The unpaired successful movement means are 24.433 for A*, 12.875 for solo and 11.436 for shared. These smaller learned-policy averages do not establish shorter routes: the methods complete different subsets of tasks. Table 4 instead holds the task fixed and retains only joint successes, where 6 of the six mean learned-minus-A* static movement differences are positive in this final experiment. Most successful single-agent learned paths tie the exact distance, while occasional long detours increase the mean gap; the failures remain visible in Table 5.

7.1 Hypothesis assessment

H1 receives directional support in 6 of six joint-success contrasts where learned mean movements exceed A*. The learned-minus-A* mean movement differences for solo with one/two/four agents are 1.462, 2.940, 5.188; shared differences are 2.185, 1.283, 0.889. Table 4 separates shorter paths and ties. A strict statement that A* always gives shorter paths is not justified by ties or any opposite-direction pairs. Static single-agent gaps have an exact reference; multi-agent prioritized paths do not. Joint-success conditioning and failures must remain visible when interpreting this conditional support for H1.

H2 is not established as a general advantage. In 3 of six contrasts the learned policy has lower joint-success post-addition completion time, while in 6 it has lower exposed-pair success than frozen A*. Learned-minus-frozen-A* completion differences for solo are -1.374, 0.723, 4.333; shared differences are -0.636, -0.669, 0.212. Table 6 reports exposed-pair success as well. No online training occurred; reacting through a fixed policy is not learned adaptation during testing. Frozen routes can wait until tick 8 removal, while replanners seek another route. Mild changes and exposure/joint-success selection prevent a broad adaptation superiority claim.

H3 is not a uniform law for every method. Across the 20 main-method adjacent agent-count/condition comparisons, 20 show lower success and 8 show increased attempts. Classical reservations can keep attempted conflicts at zero, while adding an unreachable or blocked-goal task can reduce team success independently of learned coordination. Nested endpoints match the board but add tasks; they do not isolate congestion from feasibility.

H4 shows shared-policy success advantages in 2 of six agent-count/condition summaries, disadvantages in 4, and ties in 0. The six paired seed-level differences and sample standard deviations are in Table 7, with 72 finer contrasts in the

supplement. Thus any support is condition-specific; there is no significance test or claim of consistent superiority from pooled averages alone. Shared replay, delayed team bonuses and differing realized training updates all vary together.

7.2 Mechanisms and interpretation

Current-map planners can search around newly blocked cells and reserve time-space for other agents; the learned policy can only choose from its compact current observation. The latter can alias different hidden situations and has no recurrent memory. Distribution shift also includes larger coordinate spans, longer routes and crossing endpoints. These are plausible explanations for observed patterns, not isolated causal mechanisms proved by ablations. Parameter sharing by itself does not prove cooperation or explicit information exchange.

The four-agent initial route-overlap proxy averages 2.233 intersecting route pairs for uniform endpoints and 5.317 for crossing endpoints. A route intersection has no time dimension and is not an actual collision. The endpoint arrangement therefore describes task structure rather than certifying a strict congestion ordering. Requested wall probabilities likewise differ from realized fractions recorded in each geometry.

7.3 Reproducibility and practical scope

Run metadata, scenario banks, schedules, raw CSV, first trajectories, timer repetitions, model hashes and source snapshots are archived. Analysis tables/figures are generated from immutable measured outputs, and the manuscript provenance record maps text sections and tables to those files. Replaying first trajectories checks decisions and environment behavior; it does not reproduce wall-clock timers exactly. Independent external reproduction and evaluation on public warehouse benchmarks would strengthen this controlled local study.

8. Limitations

This study uses small synthetic unit-cost grids, at most four agents and simplified synchronous moves; no real hardware validation is performed. Dynamic graphs contain only one safely scheduled addition/removal, not moving obstacles, changing costs or repeated adversarial changes. Some methods finish before exposure and some changes leave routes unaffected. Classical algorithms receive more map information than policies. A shared movement referee enforces zero actual collisions, so safety cannot be attributed to the learned networks.

Training is restricted to static 5×5 maps and a fixed budget with no final-test tuning. The observation is partially observable and the model has no memory, communication, demonstrations or graph embeddings. There are only five trained seeds and ten final geometries per size/density/arrangement stratum. Geometry rejection for boundary endpoint availability biases crossing cases, and route distance varies with the arrangement. Unreachable tasks are retained but full team solvability is not established by an exact solver.

Prioritized planning is incomplete and not globally optimal, with fixed priority and tie-breaking. Reward objectives differ from arrival-time search; waits and executed movement costs are distinct. Solo/shared training matches assigned tasks but not transition/update counts or reward timing. Path means condition on success, and matched completion means condition on joint success/exposure. Descriptive seed standard deviations are not confidence intervals. No universal speed ranking follows from timers with fixed ordering, no warm-up and shared-machine/logging effects.

The local experiment archive is provided; no public repository, independent third-party reproduction, external peer review or publication acceptance is claimed. This manuscript is a completed scientific source and readable review PDF. Venue-specific formatting and a separate reference-format Word deliverable remain the subsequent checkpoint.

9. Conclusion

The study completed a frozen comparison of classical prioritized search and DQN policy transfer across matched static/dynamic grids. Replanning A*, shared DQN and solo DQN achieved overall success rates of 89.17%, 22.19% and 24.17% respectively on their matched scenario sets. The results and paired tables answer the research question for the declared models and environments, with unsuccessful episodes retained. They do not justify a universal DQN advantage, learned collision-safety claim, or global multi-agent path optimum.

Current-map prioritized A* completed a larger fraction of matched tasks than both frozen learned policy types under this protocol. Shared two-agent training did not improve pooled success over independent deployment of the solo-trained policy.

The evidence supports evaluating reliability together with conditional path quality, separate training/inference budgets and explicit information access. Future work should add an information-matched baseline, multiple training sizes, recurrent or communicating policies, graph neural networks, expert imitation, stronger complete MAPF baselines and public warehouse tasks. PPO, QMIX and actor-critic methods are candidates for future comparisons, not methods evaluated here.

References

[1] MIT OpenCourseWare. Lecture 9: Breadth-First Search. 6.006 Introduction to Algorithms, Spring 2020, 2020. Primary source.

[2] E. W. Dijkstra. A note on two problems in connexion with graphs. Numerische Mathematik, 1, 269–271, 1959. 10.1007/BF01386390.

[3] P. E. Hart, N. J. Nilsson, and B. Raphael. A Formal Basis for the Heuristic Determination of Minimum Cost Paths. IEEE Transactions on Systems Science and Cybernetics, 4(2), 100–107, 1968. 10.1109/TSSC.1968.300136.

[4] V. Mnih et al. Human-level control through deep reinforcement learning. Nature, 518, 529–533, 2015. 10.1038/nature14236.

[5] D. Silver. Cooperative Pathfinding. Proceedings of AIIDE, 1(1), 117–122, 2005. 10.1609/aiide.v1i1.18726.

[6] R. Stern et al. Multi-Agent Pathfinding: Definitions, Variants, and Benchmarks. Proceedings of the Twelfth International Symposium on Combinatorial Search, 10(1), 151–158, 2019. 10.1609/socs.v10i1.18510.

[7] G. Sartoretti, J. Kerr, Y. Shi, G. Wagner, T. K. S. Kumar, S. Koenig, and H. Choset. PRIMAL: Pathfinding via Reinforcement and Imitation Multi-Agent Learning. arXiv:1809.03531v3, revised February 2019, 2019. 10.48550/arXiv.1809.03531.

[8] Q. Li, F. Gama, A. Ribeiro, and A. Prorok. Graph Neural Networks for Decentralized Multi-Robot Path Planning. arXiv:1912.06095, 2019. 10.48550/arXiv.1912.06095.

[9] H. Dai, E. B. Khalil, Y. Zhang, B. Dilkina, and L. Song. Learning Combinatorial Optimization Algorithms over Graphs. Advances in Neural Information Processing Systems 30, 2017. Primary source.